Firmata

Firmata is an Arduino firmware that is loaded onto the Arduino board to allow other programming languages to easily communicate with sensors, motors, and other devices. Rather than writing a custom sketch for the Arduino each time we need to communicate with a computer, like we did in our Serial-based projects in Chapter 10, we can use the standard communication protocols that have been written for us already in Firmata, along with code that is freely available for many of the more popular programming languages. Getting started with Firmata is easy enough, as the most recent versions are included with the standard Arduino distribution. There are several different versions of Firmata available, so it might take a little experimentation to find the version that works for your purposes. All we need to do is open a version of Firmata as shown in Figure 11-11; we're using the version called `StandardFirmata_2_2_forUno_0_3`. Upload this sketch to the Arduino interface board, as usual.

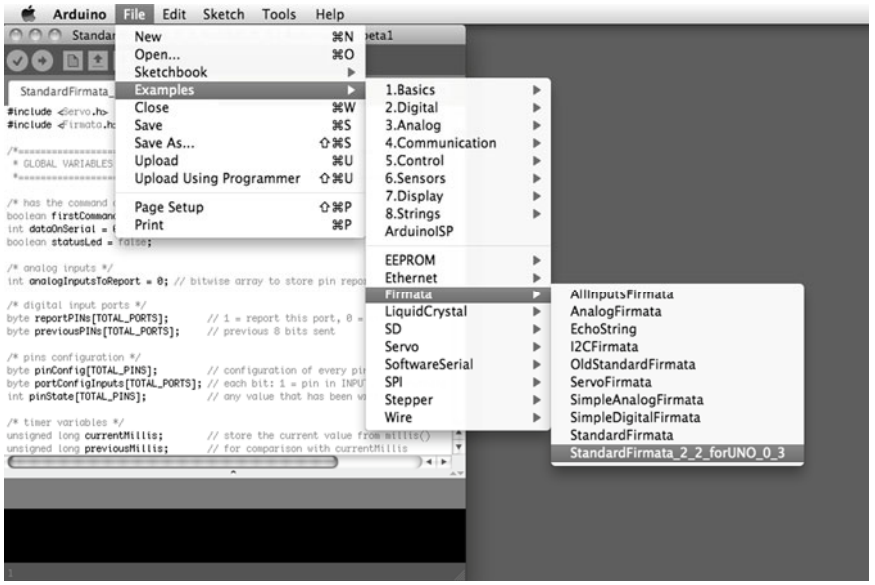


Figure 11-11. *Opening Firmata*

With Firmata loaded on the Arduino board, we need to find a matching object or test program for the language we want to use with it. We'll get to using Firmata with other languages in a moment, so just to test that everything works, we can use the Firmata Test Program, available from `http://firmata.org/wiki/Main_Page` (see Figure 11-12).